Solution

Q.1 **Express the complex number $(-\sqrt{3}+i)^2$ in the exponential from.**

Solution:

$$z=(-\sqrt{3}+i)^2$$

$$z=re^{i\theta}$$

$$z=(-\sqrt{3}+i)^2$$

$$z=(\sqrt{3})^2+i^2-2(\sqrt{3})(i)$$

$$z=3-1-2\sqrt{3}i$$

$$z=2-2\sqrt{3}i$$

$$r=\sqrt{(2)^2+(2\sqrt{3})^2}$$

$$r=\sqrt{4+4(3)}$$

$$r=\sqrt{16}$$

$$r=4$$

Now finding θ

$$\theta=\tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta=\tan^{-1}\left(\frac{-2\sqrt{3}}{2}\right)$$

$$\theta=\tan^{-1}\left(-\sqrt{3}\right)$$

$$\theta=-\frac{\pi}{3}$$

Exponential from:

$$z = 4e^{-\frac{i\pi}{3}}$$

$$z = 4\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

$$z = 4\left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)$$

Q.2 **Express the function f(x) = $\bar{z}$ Im(z) + z Re (z) + z $\bar{z}$ in form f(z) = u(x,y) +*i*v(x,y), where z = x+*i*y**

Solution: z = x + *i*y so $\bar{z}$ = x - *i*y

Re(z) = x

Im(z) = y

f(x) = $\bar{z}$ Im(z) + z Re (z) + z $\bar{z}$

f(x) = (x – *i*y)y + (x + *i*y)x + (x + *i*y) (x – *i*y)

f(x) = xy – *i*y^2 + x^2 + *i*xy + x^2 – i^2 y^2

f(x) = xy – *i*y^2 + x^2 + *i*xy + x^2 + y^2

f(x) = 2x^2 + y^2 + xy – *i*y^2 + *i*xy

u(x ,y) = 2x^2 + y^2 + xy

*i*v(x ,y) = y^2 + xy

Q.3 **Find the 4th roots of -1**

Solution: Z_0 = -1

Z_0 = -1 + *i*

So x = -1 and y = 0

$$r_0 = \sqrt{x^2 + y^2}$$
$$r_0 = \sqrt{(-1) + (0)^2}$$
$$r_0 = 1$$
$$Now\ find\ \theta$$
$$\theta = \tan^{-1}\left(\frac{x}{y}\right)$$
$$\theta = \tan^{-1}\left(\frac{0}{-1}\right)$$
$$We\ take \theta = \pi \text{ since} -1 \ lies\ on\ the\ negative\ x-axis$$

Exponential Form:

$$z_0 = r_0 e^{i\theta}$$
$$z_0 = 1e^{i\pi}$$
$$z_0 = e^{i\pi}$$

For all possible angles we use the general formula by adding the multiples of 2π

$$z_0 = e^{i(\pi+2k)}, k = 0,1,2,3$$

Finding the forth root:

$$c_k = z_0^{\frac{1}{4}}$$
$$c_k = \left(e^{i(\pi+2\pi k)}\right)^{1/4}$$
$$c_k = e^{i\frac{\pi+2\pi k}{4}}$$

For k = 0

$$c_0 = e^{i\frac{\pi}{4}} = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$$

$$c_0 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

For k = 1

$$c_1 = e^{i\frac{3\pi}{4}} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}$$

$$c_1 = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

For k = 2

$$c_2 = e^{i\frac{5\pi}{4}} = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}$$

$$c_2 = -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

For k = 3

$$c_3 = e^{i\frac{7\pi}{4}} = \cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}$$

$$c_3 = \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

For any error please contact

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